

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

Technical Presentation

Criteria	Technical Content Accuracy(3)	Problem Identification(2.5)	Use of Tools(2)	Communication(1.5)	Professionalism(1)	Total(10)
Weightage	30%	25%	20%	20%	10%	100%
Q1						
Q2						

Code of Conduct:

I _____Siva Charan (192525039)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siva Charan

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Architecture of Data Warehouse

CSA1607-DATA WAREHOUSING AND DATA MINING

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Introduction

- **What is a Data Warehouse?**
- A centralized repository for storing integrated data.
- Collects data from multiple sources.
- Stores historical and current business data.
- Supports Business Intelligence (BI).
- Helps organizations make better decisions.

Need for Data Warehouse

- **Why is it Needed?**
- Integrates data from different systems.
- Improves data consistency and quality.
- Enables faster reporting and analysis.
- Supports strategic business decisions.
- Reduces redundancy and improves efficiency.

Three-Tier Architecture

- **Three Layers**
- **Bottom Tier:** Data Sources (ERP, CRM, Databases)
- **Middle Tier:** Data Warehouse Server
- **Top Tier:** BI Tools (Reports, Dashboards, OLAP)
- Provides secure and organized data storage.
- Supports business analytics.

ETL Process



Extract

Transform

Load

- **ETL (Extract – Transform – Load)**
- **Extract:** Collect data from multiple sources.
- **Transform:** Clean and organize the data.
- **Load:** Store processed data in the warehouse.
- Improves data quality.
- Makes data ready for analysis.

Conclusion

- A **Data Warehouse** integrates data from multiple sources into a single repository.
- The **ETL process** ensures that data is accurate, clean, and consistent.
- The **three-tier architecture** enables efficient storage, management, and analysis of data.
- Data Warehouses support **Business Intelligence (BI)** and help organizations make informed decisions.